

TUDOR POPA

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Hands-on engineering leader with two decades of experience combining software architecture, people management, and platform delivery. Principal Software Engineering Manager at Microsoft, leading Fluent UI React v9 and AI-powered developer tooling. Focused on making AI useful and dependable for developers through retrieval-augmented assistants, MCP tooling, coding-agent workflows, and rigorous evaluation, backed by experience in frontend platforms, distributed systems, and cloud infrastructure.

SKILLS

- Applied AI & Agents: LLM applications, Coding agents and skills, Multi-agent workflows, Prompt and context engineering, Model Context Protocol (MCP)
- Retrieval & Knowledge Systems: Retrieval-augmented generation (RAG), GraphRAG, Knowledge graphs, Knowledge-base ingestion and indexing
- LLM & Agent Evaluation: Promptfoo, LLM-as-judge, Retrieval relevance, Model comparison, Agent benchmarking, CI-backed evaluations
- Generative UI: GenUI, Model-first design systems, Headless components, Accessibility, Design-system compliance
- Programming Languages: TypeScript, Python, JavaScript, Rust, Go
- Full-Stack Development: React, Fluent UI, Node.js, Astro, Vue
- Cloud & Delivery: Azure OpenAI, Azure, AWS, Google Cloud, Docker, Kubernetes, Terraform, GitHub Actions, Azure Pipelines
- Databases: PostgreSQL, MongoDB, Redis, Couchbase
- Machine Learning Foundations: TensorFlow, Keras, Brain.js, Predictive modeling, Genetic algorithms

WORK EXPERIENCE

Microsoft

January 2022 – Present
Prague, Czechia

Principal Software Engineering Manager

- Lead a team of 6 engineers in the Office Product Group, responsible for Fluent UI React v9 and AI-powered developer tooling; own technical strategy, delivery, hiring, and engineer development.
- Led and contributed hands-on to a GraphRAG-based developer assistant, connecting curated design-system knowledge to chat and API experiences.
- Led delivery of Model Context Protocol (MCP) tooling that grounds coding agents in component APIs, design tokens, icons, and documentation, with deterministic validation tools.
- Authored reusable coding-agent instructions and skills for Fluent UI development and review; added human-in-the-loop issue triage and ownership-aware reviewer assignment.
- Established CI-backed evaluations with Promptfoo, relevance assertions, and LLM-as-judge scoring to assess knowledge-base answer quality and guide retrieval and prompt improvements.
- Designed and contributed to GenUI benchmarks comparing coding agents and models on API correctness, accessibility, browser behavior, and design-system compliance, combining deterministic checks with LLM-based judging.
- Authored research on model-first UI generation and used benchmark findings to shape headless design-system strategy, separating accessible component behavior from styling.

- Led Fluent UI React v9 platform modernization and adoption, including React 18/19 support, server rendering, migration guidance, documentation, and recurring engineering office hours.
- Led adoption of Astro for the Fluent 2 website; the team delivered 200+ pages in 10 months and halved new-page development time (<https://astro.build/case-studies/microsoft/>).

Getir

April 2021 – December 2021

Engineering Manager

Cluj-Napoca, Romania

- Defined strategic roadmap for a team of 12 engineers, leading cross-functional collaboration and mentoring junior engineers.
- Managed and grew squads in the Algorithms Tribe, Infrastructure Alliance.
- Led infrastructure algorithms squads, delivering major initiatives to make real-time data feel real-time.
- Empowered and unblocked squads to deliver impact by establishing efficient execution and operational processes.
- Participated in and led software architecture decisions.
- Collaborated with other squads and tribes on cross-functional feature delivery.
- Planned and implemented monitoring systems to prevent service degradations and maintain high throughput.
- Guided team members in career growth, from onboarding to promotions, fostering long-term development.
- Led teams responsible for core business features, including order assignment and courier management.

Black Tree Software

Sep 2012 – April 2021

Chief Technology Officer & Co-Owner

Cluj-Napoca, Cluj, Romania

- Defined strategic roadmap for a team of 15 engineers, led cross-functional collaboration, and provided mentorship to junior engineers.
- Managed and led multiple engineering teams on various outsourcing projects.
- Drove the development of a sports betting and online gambling white-label PaaS with operations in three African countries (Ghana, Uganda, and Kenya).
- Designed and implemented the software architecture of the whole platform, creating a decoupled, highly scalable, micro-services based system.
- Worked alongside a DevOps team on designing and implementing a complete continuous integration / continuous delivery solution using the AWS cloud.
- Implemented machine learning data models for predicting and categorizing user activities / behaviors.
- Developed a lottery extraction tool which allowed the client to set a custom RTP (return to player) percentage using genetic algorithms.
- Worked with technologies ranging from low level code designed for currency detectors (ccTalk protocol) to full stack solutions (NodeJS, Python and React).

HiPPONET

Chief Technology Officer

Sep 2012 – Sep 2016
Cluj-Napoca, Cluj, Romania

- Managed a team of 6 developers on various outsourcing projects.
- Collaborated with Swedish company Penvision, developing a SaaS platform for online storage and processing of handwritten documents using proprietary handwriting recognition.
- Re-architected Rahmqvist's CRM tool for the Swedish market.
- Developed and published the Rahmqvist PIMNotes app for both Android and iOS

Wixi.com

Lead Engineer / Local Manager

Sep 2008 – Sep 2012
Cluj-Napoca, Cluj, Romania

- Managed the Cluj-Napoca office operations.
- Managed a team of 5 engineers building a file hosting service competing directly with Dropbox.
- Architected and implemented a full UI library with comparable window manager-like features in vanilla JS

OpenData

Software Engineer

Mar 2008 – Jul 2008
Cluj-Napoca, Cluj, Romania

- Designed and implemented a CRM tool for Cluj's largest real estate agency.

WebHorizon

Junior Software Engineer

Apr 2006 – Jul 2006
Cluj-Napoca, Cluj, Romania

- Worked on the Audi France website, developing components for both the client-facing site and back-office.

EDUCATION

Babeş-Bolyai University

BS Computer Science, Mathematics

Oct 2006 – May 2009
Cluj-Napoca, Cluj, Romania

Horea, Cloşca şi Crişan High School

Informatics, Mathematics

Sep 2002 – Jun 2006
Alba-Iulia, Alba, Romania

SIDE PROJECTS

AgentRig

TypeScript, MCP, LLM evaluation

<https://github.com/doidor/agentrig>

Built a CLI for sharing agent instructions and MCP configurations across coding tools, with fixture-based behavioral evaluations, separate producer/judge conversations, and baseline comparisons.

Markbook

TypeScript, React, Vite, Agent skills

<https://github.com/doidor/markbook>

Built a Markdown-first documentation and React component showcase tool with Vite, reusable coding-agent skills, and synchronized HTML and llms.txt outputs from a single source.

Github profile

Javascript, Python, Bash

<https://github.com/doidor>

Various open source projects that I personally use.

Random maze generator

Javascript

A personal project that's designed to feed a multiplayer, mobile game with random generated maps, using a genetic algorithm and ranked by an A-star algorithm score in conjunction with a complexity index.